KING SAUD UNIVERSITY
Deanship of Common First Year
Department of Basic Sciences

STAT 101
Introduction to Probability and Statistics

Semester: First
Year: 1448H
Number of Pages: 2

Tutorial (3)

Student Name: ..... **ID:** ..... **Section:** .....

Class Copy

Question 1: Select the correct answer for each of the following:

1) The lower fence LF used to identify extreme values is given by

a) $Q_1 - 1.5(Q_3 - Q_1)$ b) $Q_1 + 1.5(Q_3 - Q_1)$ c) $Q_3 - 1.5(Q_3 - Q_1)$ d) $Q_3 + 1.5(Q_3 - Q_1)$

2) The first quartile Q_1 divides the ordered data such that the percentage of values at or above it is

a) 25% b) 50% c) 75% d) 100%

3) Which of the following statistics is NOT included in the five number summary of a dataset?

a) First quartile b) Mean c) Median d) Smallest value

4) The rank of the rth percentile for an ordered set of data of size n is given by the formula

a) $\frac{r(n+1)}{100}$ b) $\frac{r(n-1)}{100}$ c) $\frac{100(r+1)}{n}$ d) $\frac{100(n+1)}{r}$

Question 2: Answer the following statements with True or False.

		Answer
1	The value of the 30th percentile can be equal to the first quartile.	.....
2	The median is equal to the 50th percentile.	.....
3	The three quartiles divide an ordered set of data into 4 equal parts.	.....
4	The fifth decile is equal to the 50th percentile, which represents the median.	.....

Question 3: Let us consider the data on the number of books read by 20 students during a semester: 3, 5, 2, 4, 3, 6, 2, 3, 4, 5, 3, 2, 4, 3, 6, 4, 3, 5, 2, 4. Then

1. Find the five numbers summary.

.....
.....
.....

2. Check if the given data has extreme values

.....
.....
.....

3. Construct the box plot for the given data.

The End– Good Luck

Student Copy

Question 1: Select the correct answer for each of the following:

1) If a dataset consists of $n = 19$ ordered observations, the rank of the 35th percentile is

a) 6.65 b) 7.00 c) 7.35 d) 6.00

2) The first quartile of a dataset is 18 and the third quartile is 42. What is the interquartile range?

a) 20 b) 24 c) 30 d) 60

3) A value exactly equal to the lower fence is

a) Always an extreme value b) Not classified as extreme c) Always the minimum d) Always Q_1

4) A dataset has $Q_1 = 25$ and $Q_3 = 45$. Which of the following values would be classified as a great extreme value?

a) 70 b) 72 c) 75 d) 80

Question 2: Answer the following statements with True or False.		**Answer**
1	Two sets of data with the same median must have the same skewness.	.....
2	Every minimum or maximum value in a dataset is automatically considered an outlier.	.....
3	A box plot displays the mean as one of its five numbers.	.....
4	The box in a box plot represents the middle 50% of the observations.	.....
5	A percentile indicates the percentage of observations that are only below a particular value	.....
6	If a student is at the 60th percentile, approximately 40% of the observations are at or above that student's value.	.....

The End– Good Luck